Code No: 136AW **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
CRYPTOGRAPHY AND NETWORK SECURITY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is an active attack?	[2]
b)	Write about substitution cipher.	[3]
c)	What is a block cipher?	[2]
d)	Enumerate the differences between private key and public key.	[3]
e)	What is message authentication?	[2]
f)	What is public key infrastructure?	[3]
g)	What is TLS?	[2]
h)	What is HTTPS?	[3]
i)	What is the need of AH in IPSec?	[2]
j)	Describe about S/MIME.	[3]

PART – B

(50 Marks)

2.a)	Explain the various types of security attacks.	
b)	Explain the model for network security.	[5+5]
	OR	
3.a)	Explain symmetric and asymmetric key cryptography.	
b)	How does steganography plays a role in network security?	[5+5]
4.	Describe in detail all steps in DES algorithm.	[10]
	OR	
5.a)	Explain the public key cryptographic principles.	
b)	Explain in detail about Diffie-Hellman Key Exchange.	[5+5]
6.	Explain the SHA – 512 algorithm.	[10]
	OR	
7.a)	List out the requirements for Kerberos.	
b)	Describe the X.509 authentication service.	[5+5]

8.a) Discuss about the web security threats and their counter measures.
b) Explain the operation of handshake protocol in SSL protocol. [5+5]

OR

9.a) Explain the challenges faced in providing security to mobile devices.
b) Describe the features of IEEE 802.11i Wireless LAN Security. [5+5]

10.a) Explain the IP security architecture.
b) List and explain the PGP services. [5+5]

OR

11. For the case study "Secure Multiparty Calculation", explore the various types of security vulnerabilities and their possible solutions. [10]

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